

Synaptic Plasticity as Short-Term Memory

One-page concept summary | DataForge 2026: Pathway Track | synaptiCITY
(github.com/sorathiyalaksh37-lang/synaptiCITY)

A standard Transformer answers a query by comparing the current token against every prior token held in a growing key-value cache: the connections between units never change, so what looks like memory is really an ever-larger lookup table rather than a property of the network's wiring. This leaves short-term memory nowhere to live except outside the connectivity itself — either in an unbounded cache, whose size scales with context length, or not at all, since the trained weights are frozen at inference and cannot represent what just happened. Synaptic plasticity proposes a different substrate: connections between units temporarily strengthen from recent co-activity, so a recently associated item becomes easier to retrieve because the pathway between two units was reweighted, not because a matching key was located by exhaustive comparison over stored history.

The classic instantiation is Hebbian learning: a synapse connecting units i and j is adjusted by $\Delta w = \eta \times a_i \times a_j$, so a connection strengthens in proportion to how often the two units fire together, and fades or is overwritten once that co-activity stops recurring. Dragon Hatchling (BDH) treats this rule as a computational mechanism rather than an analogy: Kosowski et al. (2025, arXiv:2509.26507) reformulate attention as synaptic memory, so the “read” a token performs over context is carried out by local edge weights that are written and updated as the model processes the sequence, instead of by a fixed matrix multiplication over a stored cache. What a Transformer computes by comparing queries against retained keys, BDH computes by letting local synapse-like weights accumulate and decay through Hebbian-style writes embedded directly in the forward pass.

synaptiCITY, the toy model built for this track, makes the update rule directly observable: a six-word vocabulary, an explicit weight matrix a learner can inspect cell by cell, and controls for learning rate and repetition that make destructive interference between competing associations visible in real time. This is useful for building intuition about $\Delta w = \eta \times a_i \times a_j$, but it is not BDH at any meaningful scale. BDH's reported behavior involves roughly 5% of neurons active at a given moment inside a much larger sparse, scale-free network, with reported pretraining-scaling experiments spanning 1B to 600B parameters. The toy model illustrates the same underlying Hebbian update in miniature; it makes no claim about reproducing BDH's sparsity, its scale-free connectivity, or its trained behavior at scale.

BDH-CQ (Engdahl et al., 2026, arXiv:2608.09888) extends this idea from short-term association to in-context task learning: demonstrations presented at inference continuously update the model's recurrent memory, and the query is then solved through iterative computation in a high-dimensional latent space rather than a verbalized chain of thought, with no parameter updates and no evaluation-task demonstrations used at inference. Its developers report 29.5% pass@2 on the public ARC-AGI-1 evaluation set at a computed cost of roughly \$0.00070 per task. This is a benchmark figure reported by the system's own authors, not an independently reproduced result, and it has no connection to what synaptiCITY runs: the toy app performs no ARC-AGI evaluation and cannot verify this number.

synaptiCITY's limitations are structural, not incidental. Its weight matrix is small and fully visible by design, so it cannot show sparse activation, cannot show a scale-free degree distribution among synapses, and cannot show the monosemantic-synapse behavior BDH reports at scale. The most important open question the toy model cannot answer is whether local Hebbian-style writes, at BDH's actual size and sparsity, avoid the destructive interference between competing associations that the toy model shows happens easily with only six words — that is, whether the mechanism that looks fragile in miniature becomes robust once embedded in a much larger, sparser, more structured network, or whether interference simply becomes harder to observe rather than genuinely absent.

Primary sources

Kosowski, A., Uznanski, P., Chorowski, J., Stamirowska, Z., and Bartoszkiewicz, M. “The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain.” arXiv:2509.26507.

Engdahl, B., Kosowski, A., Chorowski, J., Stamirowska, Z., Uznanski, P., et al. “BDH-CQ: In-Context Learning with Recurrent Latent Reasoning.” arXiv:2608.09888.